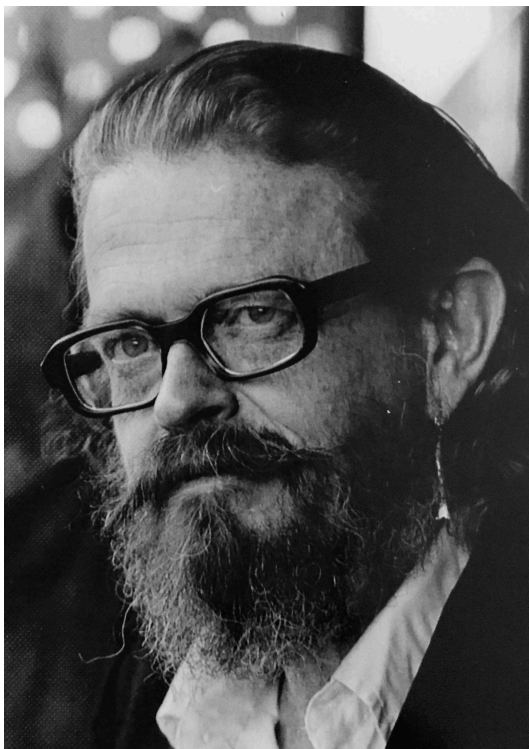


Experimental Replication of Marinov's Siberian Coliu Device

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Dedicated to the memory of Stefan Marinov and James Paul Wesley. May history prove them right and bring them the recognition they deserve.



Abstract

This document contains a detailed analysis of the so-called Marinov motor (Siberian Coliu). The device, which has been carefully rebuilt, is unexplainable from the point of view of standard Maxwellian electromagnetism. Considerable effort has been applied to ensure that the experimental conditions leave no doubt on the presence of possible artifacts that may spoil the observed phenomena and mislead the unsuspecting scientist.

An explanation of the device is provided based on the alternative formulation of electrodynamics proposed by James Paul Wesley.

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1. Wesley's theory of electrodynamics

1.1. Of frogs and total derivatives

I would like to start with an analogy. It is well-known that frogs are sensitive to the time derivative of the temperature of the fluid in which they are immersed. If one wishes to boil a frog, and throws it into boiling water; the frog will jump out of the pot. However, if one puts the frog in lukewarm water and starts slowly heating the water; the frog will not leave the fluid until it dies. This behavior can be modelled by saying that a frog jump is triggered if the frog detects a temperature change larger than a certain threshold value. In other words, if T is the temperature and t is the time, a jump is triggered if:

$$\frac{dT}{dt} > k \quad (1)$$

Let us now consider a frog that walks on a metallic surface on which there exists a spatial temperature distribution that does not change with time. For example:

$$T(x) = mx + T_0 \quad (2)$$

If the frog walks on the hot surface with speed v , the frog will experience a time varying temperature even though the temperature of the surface does not change in time. In particular, the temperature that the frog will feel can be obtained by substituting x by vt :

$$T_{frog}(t) = mvt + T_0 \quad (3)$$

And the temperature variation:

$$\frac{dT_{frog}}{dt} = mv \quad (4)$$

So we can conclude that a jump will be triggered if the following inequality is satisfied:

$$v > \frac{k}{m} \quad (5)$$

The point of this little story is that the frog does not know if the surface temperature is actually changing in time or not. All it knows is that it is getting hotter, and if it gets hot too quickly, it will jump regardless of the cause being an actual temperature change of the surface, or the movement of the frog in a region where there exists a spatially varying temperature.

Let's now consider a region of space in which there exists an electric potential and a magnetic vector potential:

$$\Phi = \Phi(x, y, z, t) \quad \vec{A} = \vec{A}(x, y, z, t) \quad (6)$$

In particular, let the scalar potential be null and the magnetic vector potential be a function of position without any dependence on time:

$$\Phi = 0 \quad \vec{A} = f(x, y, z)\vec{u}_x + g(x, y, z)\vec{u}_y + h(x, y, z)\vec{u}_z \quad (7)$$

The electric and magnetic field, according to Maxwell's equations, will be given by:

$$\vec{E} = -\vec{\nabla}\Phi - \frac{\partial \vec{A}}{\partial t} = 0 \quad , \quad \vec{B} = \vec{\nabla} \times \vec{A} \quad (8)$$

Let's focus on the electric field for now. If we consider a charged particle, with charge q , moving with speed:

$$\vec{v} = \vec{v}(x, y, z, t) \quad (9)$$

The electric force it will experience will be given by:

$$\vec{F}_e = q\vec{E} = q \left(-\vec{\nabla}\Phi - \frac{\partial \vec{A}}{\partial t} \right) = 0 \quad (10)$$

The partial derivative has a profound implication. Notice that, even though the magnetic vector potential does not change with time, since it changes with space, the particle should feel a variation as it moves, just as happened with the frog of our previous analogy.

Is the particle more intelligent than the frog? For, if we are to believe in the equation, the particle apparently is able to know the origin of the time variation of the magnetic vector potential and only experiences a force if the variation is intrinsic to the field. This clearly goes against logic. To make it even more clear, the contradiction is the following:

The following two situations are equivalent from the point of view of a test charge:

1. Remaining at rest in an area of space in which there exists a magnetic vector potential that increases linearly with time.
2. Moving with constant speed in an area of space in which there exists a time-stationary magnetic vector potential that increases linearly with a spatial coordinate.

However, according to classical electrodynamics, in case 1 an electric force will arise while in case 2 no electric force will be felt. Two different outcomes for two equivalent scenarios are a contradiction.

1.2. Historical attempts to solve the problem

According to the logic discussed in the previous section, the partial derivative of the expression of the electric field should be substituted by a total derivative:

$$\vec{E} = -\vec{\nabla}\Phi - \frac{d\vec{A}}{dt} \quad (11)$$

This idea was already considered by Hertz [1] and Pinheiro [3], who wrote:

$$\frac{d\vec{A}}{dt} = \frac{\partial \vec{A}}{\partial t} + (\vec{v} \cdot \vec{\nabla}) \vec{A} \quad (12)$$

Although this expression for the derivative makes electromagnetism galilean invariant, it fails to explain well-known experiments such as Faraday's disk, for which it predicts a wrong induced electromotive force according to the analysis of Wesley [4].

1.3. Faraday's disk

Faraday's disk is an experimental device that invalidates the hypothesis proposed by Hertz and disclosed in the previous section. Let's consider by hypothesis that Hertz was right: the magnetic field does not exist, only the electric field does, and it is defined as:

$$\vec{E} = -\vec{\nabla}\phi - \frac{d\vec{A}}{dt} = -\vec{\nabla}\phi - \frac{\partial\vec{A}}{\partial t} - (\vec{v} \cdot \vec{\nabla})\vec{A} \quad (13)$$

Let's consider now a Faraday disk like the one shown in the figure, with the conductor rotating around the axis z with angular velocity ω and the magnetic field aligned with the z axis with magnitude B_0 .

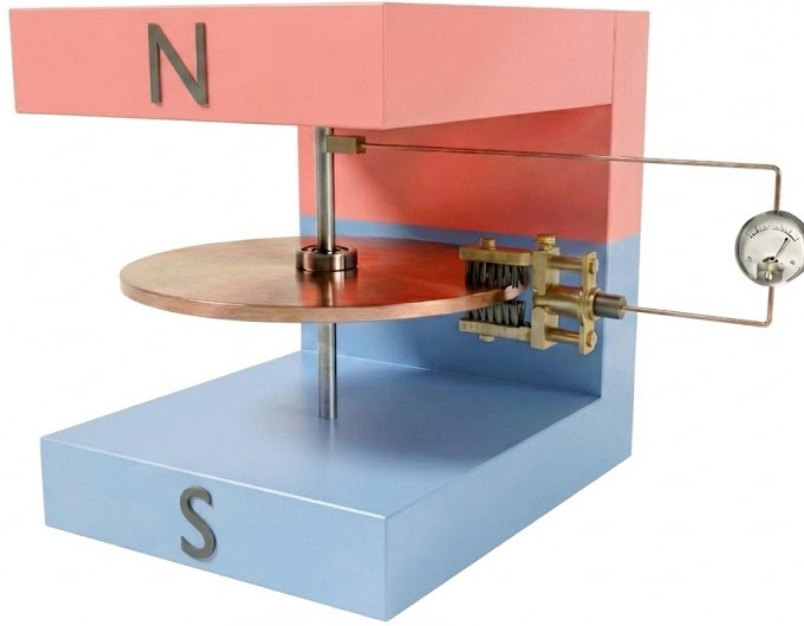


Figure 1: Faraday's disk. A copper disk rotates inside a constant magnetic field. Current is collected with brushes at the rim of the disk and at the axle.

If we consider the magnetic field to be constant in the region of the conductor, the magnetic vector potential will be given by:

$$\vec{A} = \frac{B_0\rho}{2}\vec{u}_\phi \quad (14)$$

The velocity of the charges in the conductive ring, caused by the rotation of the ring, is given by:

$$\vec{v} = \omega\rho\vec{u}_\phi \quad (15)$$

Since there are no charged bodies, the electric field is given only by the time derivative of the magnetic vector potential, with no component from the gradient of the electric potential. And since there is no explicit time variation of the magnetic vector potential, the total derivative reduces to the convective term:

$$\vec{E} = -(\vec{v} \cdot \vec{\nabla})\vec{A} = \frac{\omega B_0\rho}{2}\vec{u}_\rho \quad (16)$$

If we integrate this electric field along the radius, we obtain the emf between the center of the disk and the rim:

$$emf = \int_0^R \frac{\omega B_0 \rho}{2} d\rho = \frac{1}{4} \omega B_0 R^2 \quad (17)$$

However, it is a very well established experimental fact [] that the emf between the center of the disk and the rim in a Faraday disk is:

$$emf = \frac{1}{2} \omega B_0 R^2 \quad (18)$$

So the theory proposed by Hertz [1] and Pinheiro [3] is invalidated by the experimental facts.

James Paul Wesley [4] proposed a correction, which it turns out is of mathematical origin. In summary, Hertz, Phipps and Pinheiro were right in attempting to take the total derivative of the magnetic vector potential with respect to time, but they considered an expression for the total derivative valid only for scalar fields (according to Wesley). When the field has direction, as is the case with the magnetic vector potential, there is an additional source of time variation that must be accounted for.

1.4. Wesley's total time derivative of the magnetic vector potential

There is a missing term in the equation (12) proposed by Hertz for the total derivative of the magnetic vector potential. To see why, let's consider a charged particle q moving with uniform circular motion around the origin in the xy-plane with constant angular speed ω and radius b . The parametric equation of the trajectory of the particle is:

$$\vec{r}(t) = b \cos(\omega t) \vec{u}_x + b \sin(\omega t) \vec{u}_y \quad (19)$$

In the same region of space in which the particle is moving, there exists a constant magnetic vector potential of value:

$$\vec{A} = A_x \vec{u}_x + A_y \vec{u}_y \quad (20)$$

If we consider the particle *perception* to be associated to a moving Frenet-Serret frame, the particle experiences the magnetic vector potential as changing direction, even though the magnetic vector potential is constant.

The normal and tangent vector to the particle are:

$$\vec{u}_\rho = \cos(\omega t) \vec{u}_x + \sin(\omega t) \vec{u}_y, \quad \vec{u}_\phi = \cos(\omega t) \vec{u}_y - \sin(\omega t) \vec{u}_x \quad (21)$$

Wesley [4] claimed that the particle feels its own normal and tangent vectors as stationary, while it perceives the laboratory frame as changing directions. Accordingly, the magnetic vector potential, which is constant in the laboratory frame, is perceived by the particle as changing direction, and the rate of change is given by:

$$\frac{d\vec{A}}{dt} = A_x \frac{d\vec{u}_x}{dt} + A_y \frac{d\vec{u}_y}{dt} \quad (22)$$

Let's compute these derivatives explicitly. The laboratory frame unit vectors can be expressed as a function of the normal and tangent vectors by solving from equation (21).

$$\vec{u}_x = \cos(\omega t)\vec{u}_\rho - \sin(\omega t)\vec{u}_\phi, \quad \vec{u}_y = \cos(\omega t)\vec{u}_\phi + \sin(\omega t)\vec{u}_\rho \quad (23)$$

We can now compute the time derivatives by considering $\vec{u}_\rho$ and $\vec{u}_\phi$ as constant, as we have claimed that the particle perceives them. We have changed the signs too, as from the point of view of the particle the velocities of the unit vectors of the laboratory frame are negative.

$$\frac{d\vec{u}_x}{dt} = \omega [\sin(\omega t)\vec{u}_\rho + \cos(\omega t)\vec{u}_\phi], \quad \frac{d\vec{u}_y}{dt} = \omega [\sin(\omega t)\vec{u}_\phi - \cos(\omega t)\vec{u}_\rho] \quad (24)$$

We can now use this result in equation (22) to obtain:

$$\frac{d\vec{A}}{dt} = A_x\omega [\sin(\omega t)\vec{u}_\rho + \cos(\omega t)\vec{u}_\phi] + A_y\omega [\sin(\omega t)\vec{u}_\phi - \cos(\omega t)\vec{u}_\rho] \quad (25)$$

Rearranging terms, we obtain:

$$\frac{d\vec{A}}{dt} = \omega [A_x\sin(\omega t) - A_y\cos(\omega t)]\vec{u}_\rho + \omega [A_x\cos(\omega t) + A_y\sin(\omega t)]\vec{u}_\phi \quad (26)$$

It is easy to notice that the expressions inside the brackets correspond to the instantaneous tangential and normal components of the magnetic vector potential:

$$\frac{d\vec{A}}{dt} = \omega [A_\rho\vec{u}_\phi - A_\phi\vec{u}_\rho] \quad (27)$$

At first sight, this result that we have just obtained seems to be valid only for the special case of a particle moving with uniform circular motion around the origin. However, thanks to the theory of differential geometry of curves, the result can be easily generalized by considering that, at each instant, the particle is moving with angular velocity $\omega(t)$ around the center of the osculating circle of the trajectory.

Let's consider a charged particle q moving according to the parametric curve:

$$\vec{r}(t) = f(t)\vec{u}_x + g(t)\vec{u}_y + h(t)\vec{u}_z \quad (28)$$

The velocity of the particle at time t will be given by:

$$\vec{v}(t) = \frac{d\vec{r}}{dt} = f'(t)\vec{u}_x + g'(t)\vec{u}_y + h'(t)\vec{u}_z \quad (29)$$

And the acceleration of the particle at time t will be given by:

$$\vec{a}(t) = \frac{d\vec{v}}{dt} = \frac{d^2\vec{r}(t)}{dt^2} = f''(t)\vec{u}_x + g''(t)\vec{u}_y + h''(t)\vec{u}_z \quad (30)$$

The unit tangent vector of the Frenet-Serret frame at time t will be given by:

$$\vec{u}_t(t) = \frac{\vec{v}(t)}{\|\vec{v}(t)\|} \quad (31)$$

And the unit normal vector of the Frenet-Serret frame at time t will be given by:

$$\vec{u}_n(t) = \frac{[\vec{v}(t) \times \vec{a}(t)] \times \vec{v}(t)}{\|[\vec{v}(t) \times \vec{a}(t)] \times \vec{v}(t)\|} \quad (32)$$

The radius of the osculating circle to the curve at time t will be:

$$R(t) = \frac{\|\vec{v}(t)\|^3}{\|\vec{v}(t) \times \vec{a}(t)\|} \quad (33)$$

And the angular velocity of the particle at a time t will be:

$$\omega(t) = \frac{\|\vec{v}(t)\|}{R(t)} = \frac{\|\vec{v}(t) \times \vec{a}(t)\|}{\|\vec{v}(t)\|^2} \quad (34)$$

Now, we can finally write the general form of equation (27). Note that the signs have been changed due to $\vec{u}_n$ pointing from the position of the particle towards the center of the osculating circle, while $\vec{u}_\rho$ pointed from the center of the circle towards the position of the particle.

$$\frac{d\vec{A}}{dt} = \omega \left[(\vec{A} \cdot \vec{u}_t) \vec{u}_n - (\vec{A} \cdot \vec{u}_n) \vec{u}_t \right] \quad (35)$$

Equation (35) accounts for the perceived time variation of the magnetic vector potential caused by the change in movement of the particle. The total change, taking into account also the explicit time variation and the time variation felt by moving in a spatially varying field, the total derivative can be written as:

$$\frac{d\vec{A}}{dt} = \frac{\partial \vec{A}}{\partial t} + (\vec{v} \cdot \vec{\nabla}) \vec{A} + \omega \left[(\vec{A} \cdot \vec{u}_t) \vec{u}_n - (\vec{A} \cdot \vec{u}_n) \vec{u}_t \right] \quad (36)$$

With some abuse of notation, for the case in which the moving body is a continuous mass distribution, the above can be rewritten as:

$$\frac{d\vec{A}}{dt} = \frac{\partial \vec{A}}{\partial t} + (\vec{v} \cdot \vec{\nabla}) \vec{A} + (\vec{A} \cdot \vec{\nabla}) \vec{v} \quad (37)$$

This last equation can be rewritten with the help of a well-known vector identity as:

$$\frac{d\vec{A}}{dt} = \frac{\partial \vec{A}}{\partial t} + \vec{\nabla} (\vec{v} \cdot \vec{A}) - \vec{v} \times (\vec{\nabla} \times \vec{A}) - \vec{A} \times (\vec{\nabla} \times \vec{v}) \quad (38)$$

The last three equations (36, 37 and 38) are fully equivalent mathematically, so they can be used as best suits the situation. However, the last of them is the one that best shows the different terms of the force, including the familiar Lorentz's term which is contained in it.

1.5. Faraday's disk revisited

Let us consider again Faraday's disk, now from the viewpoint of Wesley: a cylindrical magnet rotating around its z axis with angular velocity ω . Attached to the top of the magnet there is a conductive disk that rotates together with the magnet. The brushes of a voltmeter are connected to the center of the conductive disk and to the external radius of the disk. We are interested in the emf measured in the voltmeter.

If the permanent magnet has a remanent flux density of B_0 and we consider the magnetic field as constant in the region of the conductive ring, as is usually done when studying Faraday's disk, the magnetic vector potential in cylindrical coordinates will be given by:

$$\vec{A} = \frac{B_0 \rho}{2} \vec{u}_\phi \quad (39)$$

The velocity of the charges in the conductive ring, caused by the rotation of the ring, is:

$$\vec{v} = \omega \rho \vec{u}_\phi \quad (40)$$

We will now compute each term of equation 38. Since the magnetic vector potential has no dependence with the ϕ coordinate, rotating the permanent magnet around the z axis causes no change in the magnetic vector potential. Accordingly, $\partial \vec{A} / \partial t$ is null.

The second term is given by:

$$-\vec{\nabla} (\vec{v} \cdot \vec{A}) = -\vec{\nabla} \frac{\omega B_0 \rho^2}{2} = -\omega B_0 \rho \vec{u}_\rho \quad (41)$$

The third term is given by:

$$\vec{v} \times (\vec{\nabla} \times \vec{A}) = \omega B_0 \rho \vec{u}_\rho \quad (42)$$

And finally the fourth term:

$$\vec{A} \times (\vec{\nabla} \times \vec{v}) = \omega B_0 \rho \vec{u}_\rho \quad (43)$$

The total force per unit charge is thus given by the addition of the three acting terms:

$$\vec{E} = \omega B_0 \rho \vec{u}_\rho + \omega B_0 \rho \vec{u}_\rho - \omega B_0 \rho \vec{u}_\rho = \omega B_0 \rho \vec{u}_\rho \quad (44)$$

If we integrate this electric field along the radius, we obtain the emf that we were looking for, and that coincides with the well-known result for Faraday's disk.

$$emf = \int_0^R \omega B_0 \rho d\rho = \frac{1}{2} \omega B_0 R^2 \quad (45)$$

The interesting thing about this is that we have obtained the correct result without using at any moment the ad-hoc created magnetic field or the expression of the Lorentz force. Rather, we have used only the electric field with the total derivative of the magnetic vector potential.

1.6. Wesley's description of induction

Let us consider a region of space in which there exist a charge density distribution and a current density distribution. The current density has four terms: the external current imposed as boundary condition, the induced current, the convective current (originated by charged bodies in motion) and the bound current (a macroscopic model of permanent magnets).

$$\rho = \rho(\vec{r}, t) \quad \vec{j} = \vec{j}_{ext}(\vec{r}, t) + \vec{j}_{ind}(\vec{r}, t) + \vec{v}(\vec{r}, t) \rho(\vec{r}, t) + \vec{\nabla} \times \vec{M}(\vec{r}, t) \quad (46)$$

The retarded potentials can be obtained by solving the following two uncoupled inhomogeneous wave equations (or equivalently by direct integration using the retarded times). This is Lorenz's gauge, with $A=0$ at infinity:

$$\nabla^2 \vec{A} - \mu\epsilon \frac{\partial^2 \vec{A}}{\partial t^2} = -\mu \vec{j} \quad (47)$$

$$\nabla^2 V - \mu\epsilon \frac{\partial^2 V}{\partial t^2} = -\frac{\rho}{\epsilon} \quad (48)$$

Once the potentials are known, the induction force volumetric density $\vec{E}$ is given by:

$$\vec{E} = -\vec{\nabla}\phi - \frac{d\vec{A}}{dt} = -\frac{\partial \vec{A}}{\partial t} - (\vec{v} \cdot \vec{\nabla}) \vec{A} - (\vec{A} \cdot \vec{\nabla}) \vec{v} \quad (49)$$

The induced current density is given by the constitutive equation of the material. For linear isotropic materials we can consider Ohm's law:

$$\vec{j}_{ind} = \sigma \vec{E} \quad (50)$$

The new charge distribution can be computed using the continuity equation:

$$\frac{\partial \rho}{\partial t} = -\vec{\nabla} \cdot \vec{j} \quad (51)$$

To better understand this formulation, let's consider how the transient solver of a finite elements software would approach Faraday's disk step by step:

1. Initially, the charge density and current density are known as initial condition. In this case, the only sources are the bound currents inside the permanent magnets.
2. The retarded potentials are computed solving the inhomogenous wave equations from the initial condition.
3. The induction force is computed. There appears a radial induction force.
4. The induced current is computed. If brushes are not connected, there arises a radial current as a transient response to the induction force.
5. The new charge distribution is computed using the continuity equation. There is a charge separation between the center and the rim of the disk.

Iteration continues until:

1. There are now two sources. The charge separation between the center and the rim, and the bound currents inside the permanent magnets.
2. The retarded potentials are computed solving the inhomogenous wave equations from the initial condition.
3. The induction force is computed. The force caused by the magnetic vector potential is cancelled by the force from the gradient of the scalar electric potential caused by the charge separation between the center and the rim.
4. Steady state is achieved: there is no current flow and the charge separation is kept between the center and the rim.

1.7. Ponderomotive effects

If we consider a region of space in which there exists an electric potential ϕ and a magnetic vector potential $\vec{A}$, an element with charge density ρ and current density $\vec{j}$ will be subjected to the mechanical force consistent with the previous induction force:

$$\vec{F} = -\rho \vec{\nabla} \phi - \rho \frac{\partial \vec{A}}{\partial t} - (\vec{j} \cdot \vec{\nabla}) \vec{A} - (\vec{A} \cdot \vec{\nabla}) \vec{j} \quad (52)$$

This general equation relates, in the framework of Marinov and Wesley, the laws of electromagnetism with the laws of mechanics.

1.7.1. Force between currents

Focusing on the force between two currents, using a well-known vector identity, the previous equation can be rewritten as:

$$\vec{F} = \vec{j} \times (\vec{\nabla} \times \vec{A}) + \vec{A} \times (\vec{\nabla} \times \vec{j}) - \vec{\nabla} (\vec{j} \cdot \vec{A}) = \vec{F}_L + \vec{F}_W + \vec{F}_M \quad (53)$$

As can be seen, the force has three terms, the first of which coincides with the Lorentz's force of Maxwell's theory. The other two terms represent additional forces that do not exist in Maxwell's theory, which I have tentatively named Wesley's force and Marinov's force.

The first term, Lorentz's force, requires no additional explanation as it is already very well known. If anything, it is a good sign that one of the forces that come out of our general analysis is Lorentz's force, as this allows continuity with the current understanding.

The second term, Wesley's force, can be disregarded for most steady currents. In fact, if the test current comes from an electric potential, its curl will be null due to a fundamental calculus identity that states that, if a vector field is a gradient of a scalar field, its curl is null.

$$\vec{F}_W = \vec{A} \times (\vec{\nabla} \times \vec{j}) = \vec{A} \times (\vec{\nabla} \times \sigma \vec{E}) = \vec{A} \times \sigma (\vec{\nabla} \times \vec{\nabla} V) = \vec{A} \times \vec{0} = \vec{0} \quad (54)$$

The third term, Marinov's force, is the gradient of a scalar field.

$$\vec{F}_M = -\vec{\nabla} (\vec{j} \cdot \vec{A}) \quad (55)$$

By definition, the line integral of a gradient over a closed path is null. The implication of this mathematical fact is that the net Marinov's force acting on a closed wire carrying a steady current is null. The cancellation of the term can be proved by Gauss theorem.

Let $\psi(\vec{r}) = \vec{j}(\vec{r}) \cdot \vec{A}(\vec{r})$. By the Gauss-Ostrogradsky theorem of vector calculus:

$$\int_V \vec{\nabla} (\vec{j} \cdot \vec{A}) dV = \oint_{\partial V} (\vec{j} \cdot \vec{A}) \hat{n} dS \quad (56)$$

Because the conduction current $\vec{j}$ is confined to the wire, $\vec{j} \equiv \vec{0}$ on any boundary ∂V lying in the surrounding non-conducting space outside the wire. Consequently:

$$\oint_{\partial V} (\vec{j} \cdot \vec{A}) \hat{n} dS \equiv \vec{0} \implies \int_V \vec{\nabla} (\vec{j} \cdot \vec{A}) dV \equiv \vec{0} \quad (57)$$

This is an exact theorem: Marinov's force density integrates to identically zero over any complete, closed current distribution. Consequently, the net external Newtonian force acting on any closed circuit of steady currents is strictly:

$$\vec{F}_{\text{net}} = \int_V \left[\vec{j} \times \left(\vec{\nabla} \times \vec{A} \right) \right] dV = \vec{F}_{\text{Lorentz}} \quad (58)$$

It can be concluded from the above that the total force acting on a closed circuit of steady currents will be the line integral of Lorentz's force, just as predicted by the standard Maxwell's theory. This was expected, as the net force on a closed circuit is a well established experimental fact.

One may wonder at this point if the effect of this force has ever been observed as local stresses in wires or movement in flexible circuits or circuits with mechanical degrees of freedom. And the answer is that indeed it has been observed many times in history, perhaps it is the mysterious Ampère longitudinal force responsible for wire ruptures and many other experiments.

2. Overview of Marinov's Device (Siberian Coliu)

We have until now presented the idea that the partial derivative of the magnetic vector potential should be substituted by a total derivative. This change has the advantage of logical consistency and simplification, since it avoids the recourse to the *ad-hoc* created magnetic field and Lorentz's force, and explains all electromagnetic phenomena just from the electric field.

However, as attractive as the idea may be, we have so far failed to present an *experimentum crucis*; that is, an experiment that is correctly predicted by just one of the two competing theories. Since we are talking about refuting no less than the classical theory of electrodynamics, the experiment should leave no room for doubt.

At this point is where Marinov's experimental genius perfectly complements Wesley's theoretical genius, for Marinov's device is that *experimentum crucis*. The device in question consists of a stationary toroidal permanent magnet that fully contains its own magnetic field. Now, in a plane perpendicular to the magnet, we spin a conductor and measure the induced voltage by means of sliding contacts.

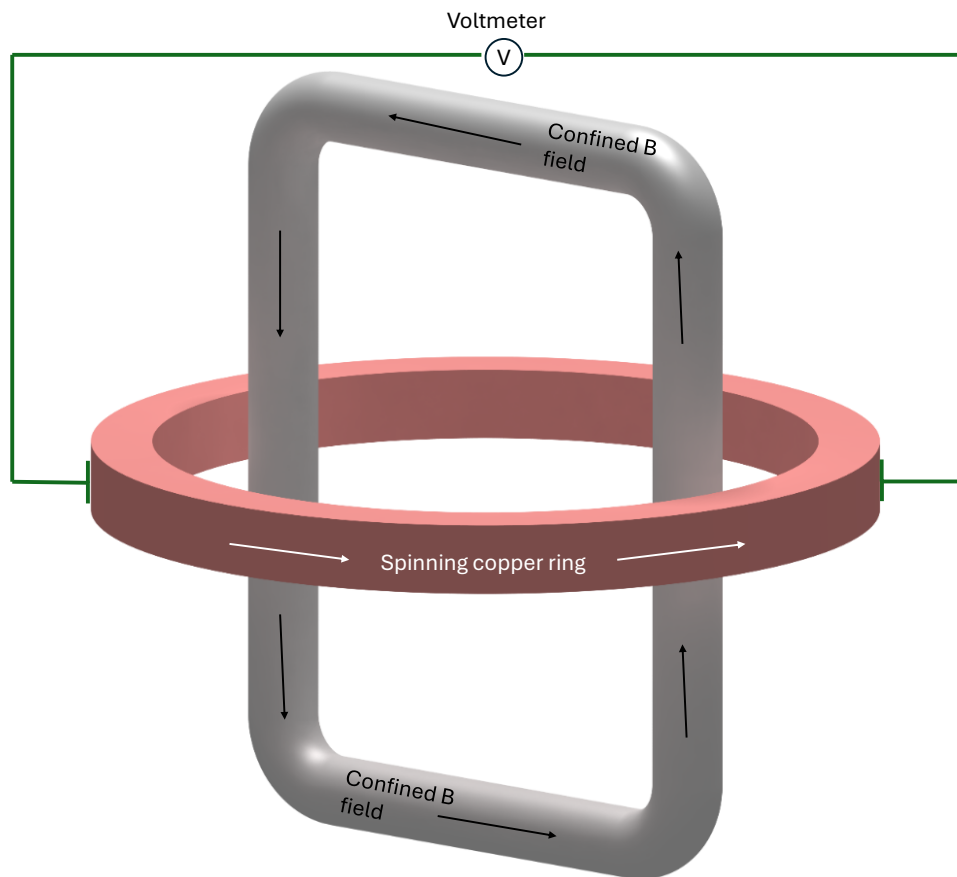


Figure 2: Concept of Marinov's device. A copper ring (shown in pink) rotates around a stationary toroidal permanent magnet (shown in grey) that contains its own magnetic flux. Two electric brushes (shown in green) are used as sliding contacts to touch the copper ring at fixed positions and measure the induced fem.

The copper ring moves in a region of space in which there is no magnetic field, since the field is confined inside the permanent magnet. For this reason, according to classical electrodynamics, no induced electromotive force is expected to be read in the voltmeter. However, as we shall see, the

permanent magnets create a spatially varying magnetic vector potential, so the movement of the copper ring should create an electromotive force according to Wesley's theory.

There is one artifact that we have to be careful about: it is reasonable to believe that a small amount of magnetic flux could leak to the area of the copper ring. This leaked flux would be in the axial direction. When the ring is forced to rotate, the drift velocity of the electrons would be in the azimuthal direction, so a Lorentz force would appear causing a radial potential difference.

Due to the special arrangement of magnets, the leakage axial magnetic flux would have opposite sign at each brush. Accordingly, at one brush, the outer radius of the copper ring would become negatively charged while at the opposite brush the outer radius of the copper ring would become positively charged, explaining the measured voltage.

If the leaked flux in the axial direction has magnitude B_L and the copper ring has inner radius R_1 and outer radius R_2 and rotates with angular velocity ω , the electromotive force measured at the brushes would be given by the formula of unipolar induction (just like in Faraday's disk):

$$emf = 2\omega \int_{R_1}^{R_2} B_L r dr \quad (59)$$

The factor of 2 comes from the fact that there are two radial potential differences connected in series, one at each brush. Since the copper ring satisfies $R_2 - R_1 \ll R_1$, we can consider B_L as constant and pull it out of the integral, obtaining:

$$emf = 2\omega B \int_{R_1}^{R_2} r dr = \omega B (R_2^2 - R_1^2) \quad (60)$$

Marinov's device can also be used as a motor instead of as a generator: when a current is supplied to the copper ring through the brushes, Marinov noted that the copper ring started rotating. However, the same undesirable artifact can take place in this situation: the supplied current is not purely azimuthal, since at the points of contact with the brushes there is a radial component.

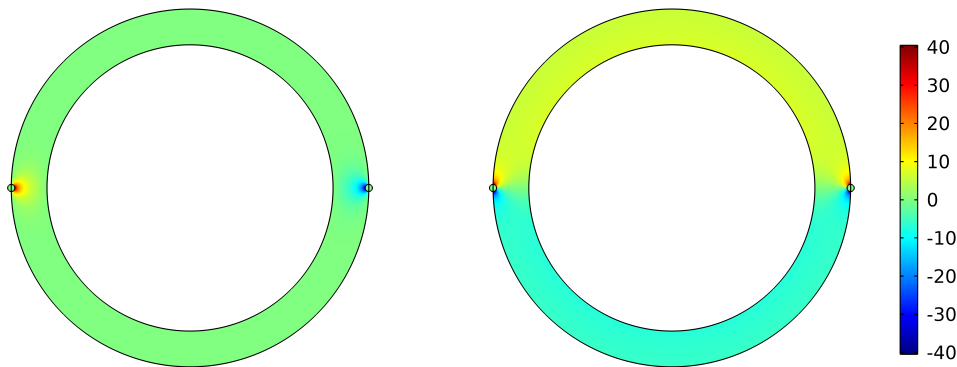


Figure 3: Radial (left) and azimuthal (right) components of the current shown as an example for an arbitrary copper ring. The result comes from solving first Poisson's equation for the voltage distribution, then computing its gradient to obtain the electric field, and finally multiplying by the conductivity to obtain the current. The units and actual values are scaled, since the example is arbitrary.

The reason for the presence of a radial component is that the current first enters perpendicularly at the points of contact with the brushes, and only after a small distance is transversed it is able to find the azimuthal trajectory.

As can be seen in the figure above, the radial component of the current has a different sign at each brush. This is obvious since the current is entering by one brush and leaving by another. Unfortunately, the leaked magnetic flux also has a different sign at each brush, allowing the possibility of a net torque caused by Lorentz's force.

Lorentz's force can thus be used to explain both the measured voltage when the device acts as generator and the net torque when the device acts as a motor. We do not know if Marinov noticed this fact, so our main priority is to discern which of the following two options is true:

1. All effects in Marinov's device are caused by Lorentz's force, which is present due to the leaked magnetic flux from the closed magnetic circuit. The device has no interest.
2. Some effects in Marinov's device can not be justified by Lorentz's force from the leaked magnetic flux, and other forces must be considered to explain the observed behavior.

To discern between the two options, we will follow two strategies: short-circuiting the radial dimension of the copper ring with the brushes and using a copper ring with a very small radial dimension ($R_2 - R_1 \approx 0.1 \text{ mm}$). Our aim, in the generator version, is to obtain an induced voltage several orders of magnitude bigger than the voltage explainable by Lorentz's force.

Leaving aside Marinov's claim of having observed a new electromagnetic force, that Wesley explained by his theory of the total derivative, Marinov made even more impressive claims for his device, namely:

1. When the device is used as a motor, there is no back electromotive force caused by Lenz's law. Rather, the only power consumption is the heat dissipated by Joule's effect.
2. When the device is used as a motor, angular momentum is not conserved. The stator does not experience the opposite torque of the rotor.
3. When the device is used as a generator, if the rotor is brought to a certain rotational speed by external means, and is then left to rotate by itself; the time required for its complete detention under friction is smaller when a larger current is drawn from the output brushes (anti-lenz effect).

We will check all these claims experimentally. But first, to be able to predict the results of Wesley's theory, we need to calculate the magnetic vector potential spatial distribution of our experimental device. To this goal we will devote the next section of this document.

3. Magnetic vector potential of Marinov's Device

3.1. Calculation by analytical means

Although we presented before the conceptual version of Marinov's device, the stator of the practical version is built in a slightly different way, with two cylindrical permanent magnets and two blocks of high magnetic permeability alloy to close the magnetic circuit as shown below.

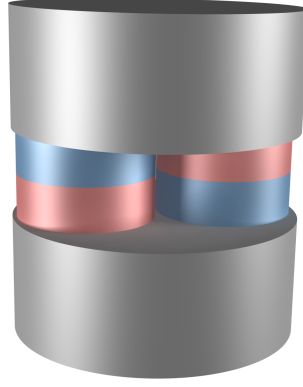


Figure 4: Marinov's stator with north poles of permanent magnets shown in red, south poles of permanent magnets shown in blue and high permeability alloy shown in gray.

3.1.1. Ideal approximation as infinitely long permanent magnets

The magnetic vector potential in the xy -plane (the horizontal plane that cuts both magnets in half) can be calculated by approximating the magnetic circuit as two infinitely long permanent magnets. This is valid as long as the magnetic flux outside the permanent magnets can be neglected against the magnetic flux inside them, as is the case in our experimental build. Accordingly, we can consider two infinitely long permanent magnets, M_1 and M_2 , each of them with radius R , both oriented parallel to $\hat{z}$, one of them magnetized with remanent flux density $B_0 \hat{z}$ and the other magnetized with remanent flux density $-B_0 \hat{z}$. Magnet M_1 crosses the xy -plane at $x = 0$, $y = b$ while magnet M_2 crosses the xy -plane at $x = 0$, $y = -b$.

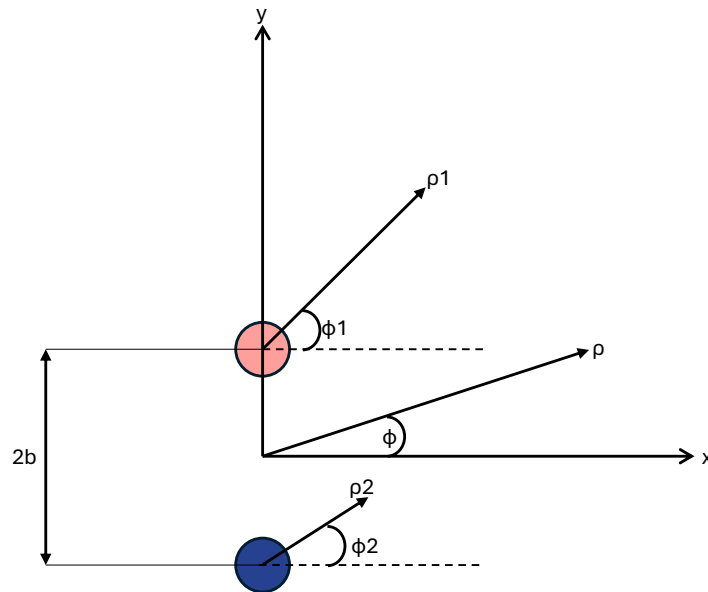


Figure 5: Idealization of the stator as two infinitely long permanent magnets.

Before solving this ideal situation, we will start with an even simpler case: the magnetic vector potential of a single infinitely long cylindrical permanent magnet of radius R and with its axis of revolution aligned with $\hat{z}$. After solving this simpler case, we will come back to the full problem.

3.1.2. Solution for a single ideal infinite cylindrical permanent magnet

Let's first consider the case of a single infinitely long cylindrical permanent magnet of radius R and remanent flux density $B_0\hat{z}$ and with its axis of revolution aligned with $\hat{z}$. After solving this simpler case, we will come back to the full problem.

The flux density outside the permanent magnet is null (because the length is infinite, just as with the case of an infinitely long solenoid) while the flux density inside it is just the remanent flux density, $B_0\hat{z}$, which is a parameter of the problem description.

If we consider a circle in the xy -plane with radius ρ , the following equation holds true for the magnetic vector potential (and for any well behaved vector field, as stated in Stoke's theorem):

$$\oint \vec{A} \cdot d\vec{l} = \iint (\vec{\nabla} \times \vec{A}) \cdot d\vec{S} \quad (61)$$

Now we can apply the definition of magnetic field, $\vec{B} = \vec{\nabla} \times \vec{A}$, to obtain:

$$\oint \vec{A} \cdot d\vec{l} = \iint \vec{B} \cdot d\vec{S} = \Phi_{enc} \quad (62)$$

The above equation means that the circulation of the magnetic vector potential equals the enclosed magnetic flux, just like the circulation of the magnetic flux density equals the enclosed current. This allows us to write the following trivial proposal for a magnetic vector potential:

$$\vec{A} = \begin{cases} \frac{B_0\rho}{2} \hat{\phi} & \rho \leq R \\ \frac{B_0R^2}{2\rho} \hat{\phi} & \rho > R \end{cases} \quad (63)$$

We will do 4 checks on the proposed potential before deeming it valid:

1. The potential must be continuous.
2. The potential must be null at infinite distance from the source.
3. The divergence of the potential must be identically null.
4. The curl of the potential must be B_0 inside the magnet and null outside it.

The first check is trivial. It suffices to check the continuity at $\rho = R$, which indeed is verified. The second check is also trivial, since the magnetic vector potential decays as $1/\rho$. For checks 3 and 4, we must first remember the expression of $\vec{\nabla}$ in cylindrical coordinates:

$$\vec{\nabla} = \frac{\partial(\rho\cdot)}{\rho\partial\rho}\hat{\rho} + \frac{\partial(\cdot)}{\rho\partial\phi}\hat{\phi} + \frac{\partial(\cdot)}{\partial z}\hat{z} \quad (64)$$

Computing $\vec{\nabla} \cdot \vec{A}$ and $\vec{\nabla} \times \vec{A}$ in both regions of equation 63 allows us to satisfy checks 3 and 4 explicitly; so our proposed potential is accepted as valid.

Now that we know the magnetic vector potential of an infinitely long permanent magnet, we can apply trigonometry to solve the case described in figure 4. We will use the coordinate systems defined in figure 5.

3.1.3. Solution of the full problem

Figure 5 shows the xy plane. The red and blue colour represent the $\hat{z}$ component of the magnetic flux density $\vec{B}$, which is positive (red) inside one magnet, negative (blue) inside the other magnet and zero (white) everywhere else.

Any point of the xy -plane can be expressed using the centered cylindrical coordinates:

$$P = (\rho, \phi) \quad (65)$$

The distance ρ_1 from point P to the axis of the magnet M_1 situated at $y = b$ will be given by:

$$\rho_1 = \sqrt{(\rho \cos \phi)^2 + (\rho \sin \phi - b)^2} = \sqrt{\rho^2 + b^2 - 2b\rho \sin \phi} \quad (66)$$

Similarly, the distance ρ_2 from point P to the axis of the magnet M_2 situated at $y = -b$ will be given by:

$$\rho_2 = \sqrt{(\rho \cos \phi)^2 + (\rho \sin \phi + b)^2} = \sqrt{\rho^2 + b^2 + 2b\rho \sin \phi} \quad (67)$$

The angles in the coordinate system of each magnet can also be expressed in terms of the angle in the global coordinate system, according to:

$$\phi_1 = \text{atan} \left(\frac{\rho \sin \phi - b}{\rho \cos \phi} \right) \quad (68)$$

$$\phi_2 = \text{atan} \left(\frac{\rho \sin \phi + b}{\rho \cos \phi} \right) \quad (69)$$

If we directly apply equation 63, the magnetic vector potential outside the permanent magnets will be given by:

$$\vec{A} = \frac{B_0 R^2}{2\rho_1} \hat{\phi}_1 - \frac{B_0 R^2}{2\rho_2} \hat{\phi}_2 \quad (70)$$

Where the minus sign in the second term of the right side is due to the magnetic flux going in the $-\hat{z}$ direction inside the second magnet, and equation 63 has been applied.

Unfortunately, this expression is in terms of $\hat{\phi}_1$ and $\hat{\phi}_2$, and we need it in terms of $\hat{\rho}$ and $\hat{\phi}$. So now the task is expressing the unit vectors $\hat{\phi}_1$ and $\hat{\phi}_2$ as a function of the unit vectors $\hat{\rho}$ and $\hat{\phi}$. The coordinate transformation is as follows:

$$\begin{aligned} \vec{A} = & \left(\frac{B_0 R^2}{2\rho_1} \sin(\phi - \phi_1) - \frac{B_0 R^2}{2\rho_2} \sin(\phi - \phi_2) \right) \hat{\rho} + \\ & \left(\frac{B_0 R^2}{2\rho_1} \cos(\phi - \phi_1) - \frac{B_0 R^2}{2\rho_2} \cos(\phi - \phi_2) \right) \hat{\phi} \end{aligned} \quad (71)$$

With much algebraic manipulation it is possible to arrive to a simplified formula:

$$\vec{A} = \frac{B_0 R^2 b (\rho^2 - b^2) \sin \phi}{(\rho^2 + b^2)^2 - 4b^2 \rho^2 \sin^2 \phi} \hat{\phi} + \frac{B_0 R^2 b (\rho^2 + b^2) \cos \phi}{(\rho^2 + b^2)^2 - 4b^2 \rho^2 \sin^2 \phi} \hat{\rho} \quad (72)$$

The full simplification is not shown, but it is easy to check that the two equations (71 and 72) are identical by giving values to the variables or by plotting.

3.1.4. Solution of a particular case

If we consider our experimental device, we can implement equation 71 in a Matlab script (which also runs in Octave) to plot the magnetic vector potential as a function of the angle in the region of the conductive ring. We have used the convention of figure 5.

```
clear; clc; close all;
B0=1.44; % Remanent flux density of each magnet [T]
R=35; % Radius of each magnet [mm]
rho=73; % Radius of the conductive ring [mm]
b=35.3; % Half distance between the magnet centers [mm]
phi=linspace(0,2*pi,500); % Angles to evaluate [rad]
x1=rho*cos(phi); x2=x1;
y1=rho*sin(phi)-b; y2=rho*sin(phi)+b;
rho1=sqrt(x1.^2+y1.^2); rho2=sqrt(x2.^2+y2.^2);
phi1=atan2(y1,x1); phi2=atan2(y2,x2);
theta1=phi-phi1; theta2=phi-phi2;
Aphi1=B0*R^2./(2*rho1); Aphi2=B0*R^2./(2*rho2);
Aphi=Aphi1.*cos(theta1)-Aphi2.*cos(theta2);
Arho=Aphi1.*sin(theta1)-Aphi2.*sin(theta2);
plot(phi/pi,Arho,phi/pi,Aphi);
grid on; legend('Ar','Ap');
xlabel('Phi (pi rad)');
ylabel('Magnetic vector potential (Wb/km)');
```

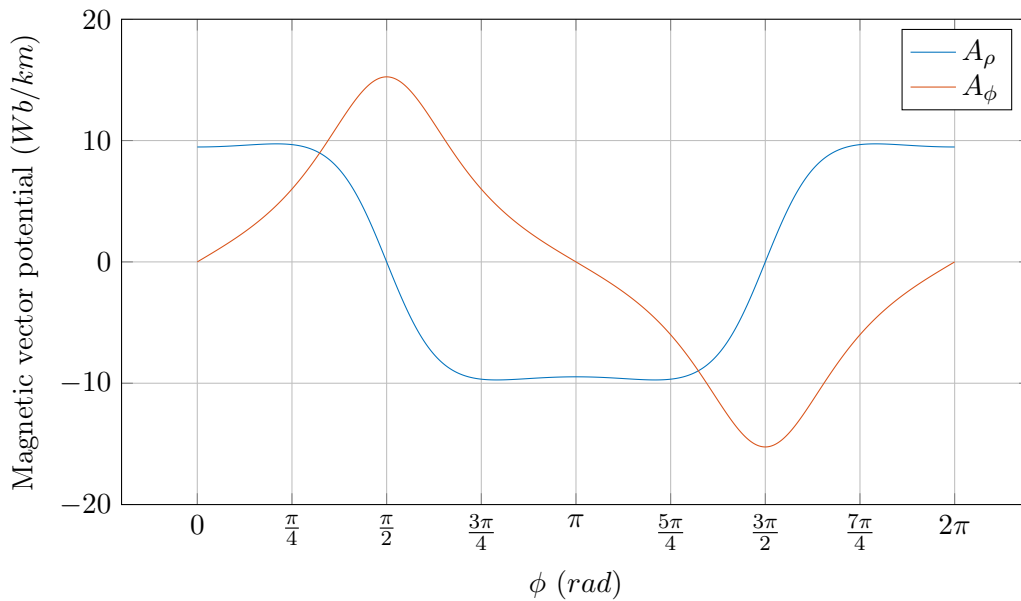


Figure 6: Magnetic vector potential in our experimental device, computed analytically.

While the above plot shows the magnetic vector potential as a function of the angle in the region of the conductive ring, we can get a better intuition of the shape of the potential by looking at a color plot of the xy-plane (the horizontal plane that cuts both magnets in half in figure 5) outside the magnets.

Please note that, in figure 7, only the areas outside the permanent magnets have been plotted, since we are not concerned with the magnetic vector potential inside the permanent magnets.

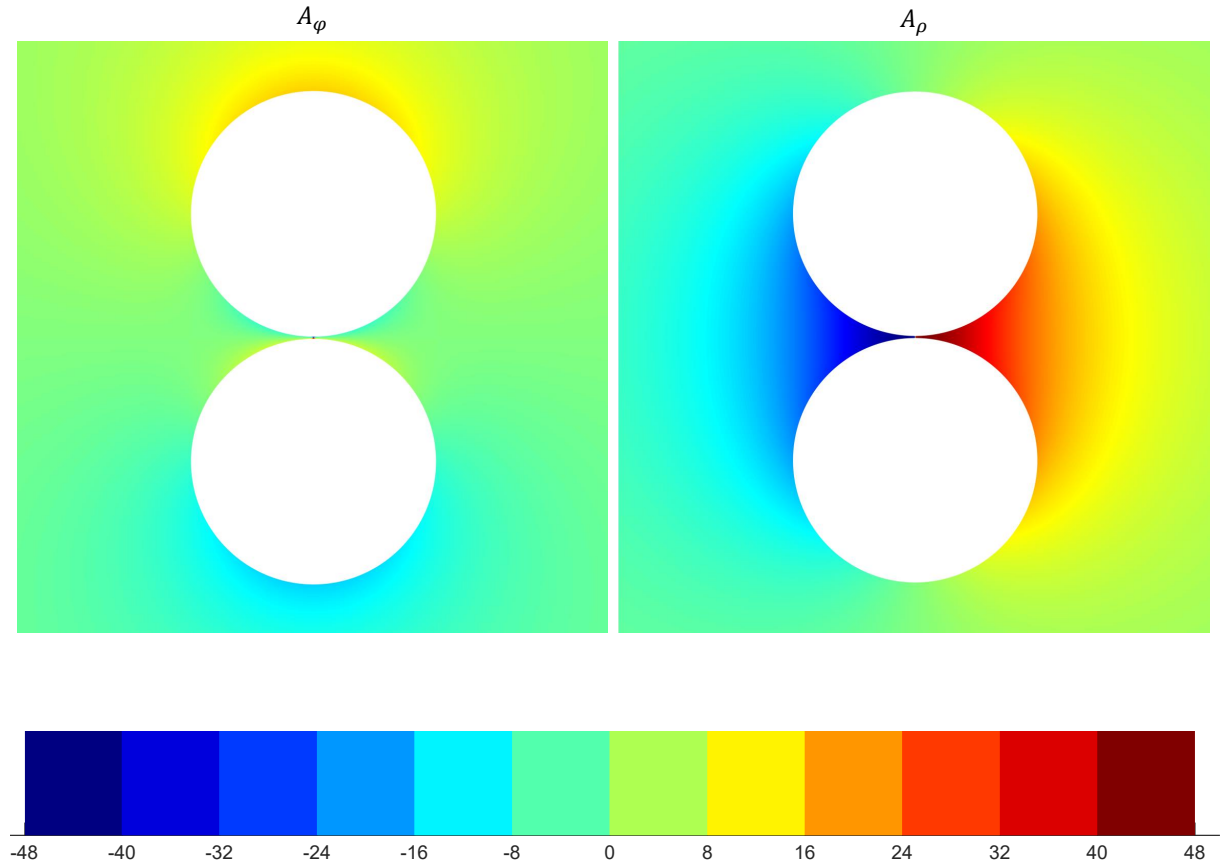


Figure 7: Color plot of the magnetic vector potential of our experimental build, computed analytically with equation 71 and expressed in Wb/km.

3.2. Calculation by finite elements

We have built a finite elements model of the stator of figure 4 in the software Comsol Multiphysics, obtaining the result shown in the following plot.

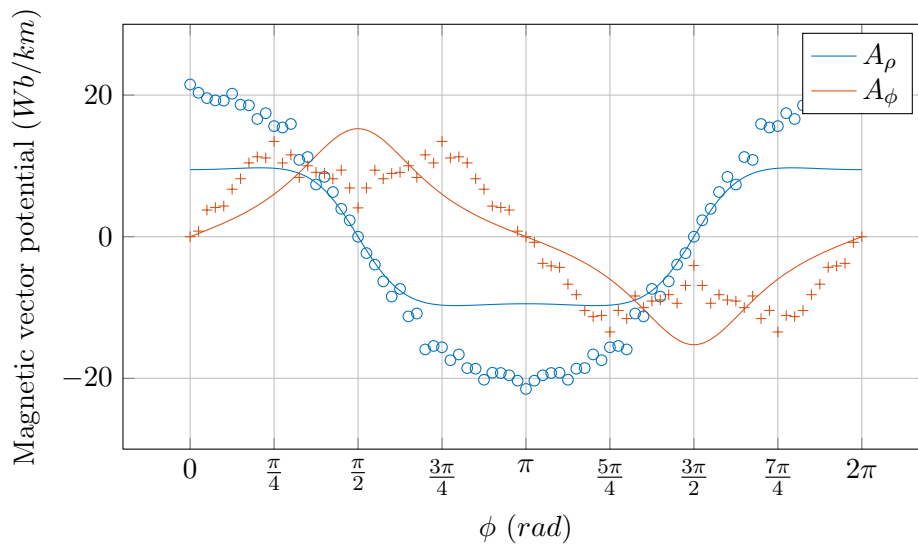


Figure 8: Magnetic vector potential in our experimental device, computed by fem. The continuous lines are the analytical solution of the ideal case, while the discrete points are obtained from the fem model.

As can be noticed, even though the order of magnitude and overall shape of the functions are the same, there are some discrepancies between the fem result and the analytical solution. However, we do not consider this a shortcoming of either solution, but rather we believe that the discrepancy is due to the departure from ideality that is taken into account in the fem model. According to this, the fem model is the actual correct solution to the problem.

These departures from the ideal case studied analytically would be, among others: the magnetic saturation of the steel blocks, the shape of the steel blocks, the magnetic flux leakage and the non-linear magnetic permeability.

The reader may be wondering why the plots give the magnetic vector potential at different azimuthal positions, but at just one radial position. The reason is that the radial dimension of the conductive ring is negligible. If R_1 and R_2 are, respectively, the inner and outer radius of the conductive ring; our experimental build satisfies:

$$R_2 - R_1 \approx 0.1 \text{ mm} \tag{73}$$

Due to this feature we are able to assume that, for a given angular position, the magnetic vector potential does not change along the radial dimension of the conductive ring. This approximation, which we deem as fully valid, will be used extensively also in the next chapter.

4. Expected results with Marinov's device

4.1. Expected induced electromotive force in generator mode

If the copper ring is forced to rotate by an external motor while the magnetic circuit is held fix, according to standard maxwellian electrodynamics, no emf should appear at the brushes. The magnetic field is zero (except for a small flux leakage, that we discussed in section 2). As for the standard electric field, it is given by:

$$E_\phi = -\nabla\Phi - \frac{\partial A_\phi}{\partial t} \quad (74)$$

There are no charged bodies and, since the the magnetic circuit is stationary, the partial time derivative of the magnetic vector potential is null in all space. The electric field is thus also null in the region of the copper ring, and no electromotive force is to be expected between the brushes. Now let's move on to Wesley's explanation: according to him, induction happens locally and is given by:

$$\frac{\vec{F}}{q} = \vec{E} = -\nabla\Phi - \frac{d\vec{A}}{dt} \quad (75)$$

The presence of a total time derivative, unlike Maxwell's partial derivative, gives rise to extra possibilities for causing induction, as we saw in previous sections. The total derivative of a well-behaved vector field $\vec{A}$, as seen by a particle moving with velocity $\vec{v}$ relative to the source of the vector field, neglecting retarded effects by assuming small velocities, was shown in equation 38, reproduced below for the comfort of the reader:

$$\frac{d\vec{A}}{dt} = \frac{\partial\vec{A}}{\partial t} + \vec{\nabla}(\vec{v} \cdot \vec{A}) - \vec{v} \times (\vec{\nabla} \times \vec{A}) - \vec{A} \times (\vec{\nabla} \times \vec{v}) \quad (76)$$

According to section 3 equation 72, the magnetic vector potential in the area of the copper ring is:

$$\vec{A} = \frac{B_0 R^2 b (\rho^2 - b^2) \sin\phi}{(\rho^2 + b^2)^2 - 4b^2 \rho^2 \sin^2\phi} \hat{\phi} + \frac{B_0 R^2 b (\rho^2 + b^2) \cos\phi}{(\rho^2 + b^2)^2 - 4b^2 \rho^2 \sin^2\phi} \hat{\rho} = A_\rho \hat{\rho} + A_\phi \hat{\phi} \quad (77)$$

The velocity field of the copper ring, in cylindrical coordinates, will be given by:

$$\vec{v} = \omega \rho \vec{u}_\phi \quad (78)$$

Let's compute now the terms of equation (38) one by one: first, since the magnetic circuit is stationary, the partial derivative of the magnetic vector potential is null.

$$\frac{\partial\vec{A}}{\partial t} = 0 \quad (79)$$

The next term is given by:

$$\vec{\nabla}(\vec{v} \cdot \vec{A}) = \vec{\nabla}(\omega \rho A_\phi) = \left(\omega A_\phi + \omega \rho \frac{\partial A_\phi}{\partial \rho} \right) \vec{u}_\rho + \omega \frac{\partial A_\phi}{\partial \phi} \vec{u}_\phi \quad (80)$$

The next term is given by:

$$-\vec{v} \times (\vec{\nabla} \times \vec{A}) = 0 \quad (81)$$

The term is null because, as we saw, there is no magnetic field outside the permanent magnets; so the magnetic vector potential is curl-free. Indeed, this term represents Lorentz's force, and the whole experiment is arranged to avoid it.

And the final term is given by:

$$-\vec{A} \times (\vec{\nabla} \times \vec{v}) = 2\omega A_\rho \vec{u}_\phi - 2\omega A_\phi \vec{u}_\rho \quad (82)$$

Applying now equation (75), we can write the azimuthal component of the electric field as:

$$E_\phi = -\omega \left(\frac{\partial A_\phi}{\partial \phi} + 2A_\rho \right) \quad (83)$$

To obtain the emf between the two brushes, which are situated at radius ρ and angles ϕ_1 and ϕ_2 , we must do the integral:

$$emf = -\rho \int_{\phi_1}^{\phi_2} E_\phi d\phi = \omega \rho \Delta A_\phi + 2\omega \rho \int_{\phi_1}^{\phi_2} A_\rho d\phi \quad (84)$$

5. Design and manufacture of Marinov's Device

5.1. Magnetic circuit

The main component of the device is what Marinov called *Nikolaev's magnet*, a magnetic circuit with a completely self-contained magnetic flux.

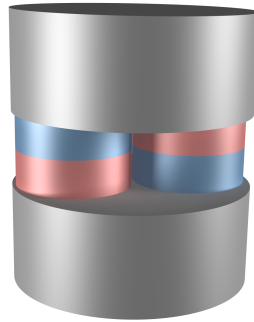


Figure 9: Closed magnetic circuit with north poles of permanent magnets shown in red, south poles of permanent magnets shown in blue and high permeability alloy shown in gray.

We have used two cylindrical neodymium magnets of 70 mm of diameter and 45 mm of height each; and two cylindrical blocks of soft steel of 140 mm of diameter and 60 mm of height each. In particular, the materials used have been neodymium magnetized to rating N52 and low-carbon steel type S355J2 [2]. The magnets have been bought in the following website (linked in october 2026): <https://www.superimanes.com/imanes-de-neodimio/discos/iman-neodimio-disco-70x45-mm>

The blocks of soft steel serve the purpose of closing the magnetic circuit by providing a low reluctance path that minimizes the flux leakage to the area of the copper ring. Due to the project being self-financed by the authors, we have been unable to buy a special alloy such as Permalloy, Mumetal or Supermalloy. If the experiment is repeated by an academic research team, we suggest that a special alloy with larger magnetic permeability should be used.

The assembling of the magnetic circuit has required the use of special wooden tooling, since the extreme force of the magnets prevented their installation by hand.

5.2. Motor

The copper ring is spun by means of an external motor. In particular, we have used a squirrel cage monophasic induction motor provided by Adajusa under reference SY-56M1-2-B5. The rotation speed is fixed at 2740 rpm, and it can provide up to 90W of power.



Figure 10: Squirrel cage motor for spinning the copper ring, provided by company Adajusa.

5.3. Conductive ring

For the conductive ring, our main concern has been making the radial dimension as small as possible to minimize the interaction between the conductive ring and the leaked magnetic flux. With this idea in mind, we have made a plastic part by 3D printing, and have attached to it a very thin layer of metallic film.

To ensure that the metallic film has a known thickness, we have bought a 0.1 mm thick foil from a provider specialized in precision gauge tape:

<https://www.hasberg-schneider.de/en/services/precision-thickness-gauge-strip-standard-dimensions/>

5.4. Assembling

A CAD model of the assembled machine can be found in the following link:

https://alberto940.github.io/Siberian_Coliu_Device/Generator.html

Anyone that wishes to examine the machine or is invited to my house to study it in detail, or to ask me any questions. My contact details are in the first page of this document.

6. Experimental results

7. Bibliography

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8. Acknowledgments

- Christof Baumgäertel, who has held the most interesting discussions and helped immensely with the interpretation of the results.
- Gabriele Wesley, the wife of the late scientist James Paul Wesley; who has kindly shared with us all books left by her husband. Her kindness and her hope in bringing the work of her husband to academic circles has been an inspiration for us.
- Javier Pérez Fernández and Luis Coll Izquierdo, who have helped us with the manufacturing of some of the parts.